

the data for a month before the failure. The system lacked the mechanism to learn from it.

Three specific limitations define the boundary of threshold-based alerting:

Point-in-time blindness: Alerts evaluate instantaneous or short-window metrics. They cannot distinguish a temporary spike from a sustained growth trajectory.

Single-dimension isolation: Alert rules evaluate individual metrics in isolation. They cannot model the compound risk of multiple metrics simultaneously trending towards saturation.

Static decay: Alert thresholds are set once and age poorly. A threshold calibrated for a workload six months ago may be dangerously misaligned with the workload today.

Machine learning addresses all three directly.

Architecting for behaviour, not rules

The framework given here is grounded in three design decisions that distinguish it from conventional autoscaling approaches.

Behavioural learning over rule definition: Rather than encoding human judgment into threshold rules, the framework learns decision boundaries from labelled historical outcomes. The model is not told that CPU above 85% means Scale Up — it observes thousands of instances where that metric pattern historically preceded degraded performance and infers the relationship. This makes the model's judgment more accurate and adaptive than any rule set a human could maintain.

The four-week observability window: A rolling four-week window of weekly metric aggregates is the core input structure. This horizon is long enough to capture genuine trends and short-term seasonal patterns, short enough to remain operationally relevant for current workload characteristics, and naturally aligned with the monthly cadence at which infrastructure planning cycles operate. Instantaneous or daily readings introduce noise; annual history introduces irrelevance.

Three-state classification: Every infrastructure instance is classified weekly into exactly one of three states as given in Table 1.

This constrained output space is a deliberate design choice. Continuous scaling recommendations introduce instability and operational overhead. A weekly, three-state classification maps directly to human decision-making cadence and produces actionable outputs.

Bootstrapping heuristics vs machine learning labels

A common pitfall in ML-driven infrastructure is training a model strictly on static rules (e.g., labelling training data as 'Scale Up' simply because CPU crossed 85%). Doing so

Classification	Condition	Action
Scale Down	Resources materially under-utilised across all four weeks	Right-size to reduce cost (schedule during low-traffic window)
Remain	Metrics stable or trending safely within headroom	No change — instance appropriately provisioned
Scale Up	Consistent upward trajectory across multiple dimensions	Proactive scaling before saturation and incident

Table 1: Three-state classification output — condition and recommended action for each state

creates a 'surrogate model' that merely memorises human heuristics rather than discovering true failure patterns. To avoid this, the framework uses a two-phased approach.

Phase 1: Operational bootstrapping: For initial deployment, the system can be seeded with decision boundaries derived from extensive operational experience:

- **Scale Down:** e.g., CPU below 40% and memory below 65% across all four weeks.
- **Remain:** e.g., Stable metrics, or upward trends within otherwise safe ranges (applying a conservative bias to prevent premature scaling).
- **Scale Up:** e.g., CPU consistently above 85% or connections consistently at or above 95% of the configured maximum.

Phase 2: True behavioural learning: As the system matures, the true machine learning labels are instead derived from actual system degradation events—such as query latency spikes, application-side connection timeouts, or OOM kills. The Gradient Boosting classifier links the preceding four weeks of telemetry to these actual historical outcomes.

 **Note:** Edge cases receive explicit handling. Weeks in which metrics read zero — indicating scheduled downtime or full inactivity — are held neutral and excluded from trend calculations.

Defining the 16-dimensional observability matrix

A single vital sign tells you little. A doctor diagnosing a patient doesn't look at heart rate alone — they read pulse, blood pressure, oxygen saturation, and temperature together, in context. Infrastructure health is no different. The model is trained not on four metrics individually, but on their joint behaviour — 16 values that together form a behavioural fingerprint no threshold rule can replicate. Four core operational metrics form the input dimensions for each classification decision: